Review Questions

The answers to the chapter review questions can be found in the Appendix.

1. Which of the following Java operators can be used with `boolean` variables? (Choose all that apply.)
   - **A.** `==` compara dos booleanos
   - **B.** `+`
   - **C.** `--`
   - **D.** `!` devuelve el valor inverso
   - **E.** `%`
   - **F.** `~`
   - **G.** Cast with `(boolean)` es un cast redundante

2. What data type (or types) will allow the following code snippet to compile? (Choose all that apply.)

```
byte apples = 5;
short oranges = 10;
_____ bananas = apples + oranges;
```

   - **A.** `int`
   - **B.** `long` La suma de byte y short devuelve un int, así que en int, long y double son correctos
   - **C.** `boolean`
   - **D.** `double`
   - **E.** `short`
   - **F.** `byte`

3. What change, when applied independently, would allow the following code snippet to compile? (Choose all that apply.)

```
3: long ear = 10;
4: int hearing = 2 * ear;
```

La operación 2 * ear da un long, por lo que se necesitar castear a int o cambiar a long

   - **A.** No change; it compiles as is.
   - **B.** Cast `ear` on line 4 to `int`.
   - **C.** Change the data type of `ear` on line 3 to `short`.
   - **D.** Cast `2 * ear` on line 4 to `int`.
   - **E.** Change the data type of `hearing` on line 4 to `short`.
   - **F.** Change the data type of `hearing` on line 4 to `long`.

4. What is the output of the following code snippet?

```
3: boolean canine = true, wolf = true;
4: int teeth = 20;
5: canine = (teeth != 10) ^ (wolf=false);
6: System.out.println(canine+", "+teeth+", "+wolf);
```

A. `true, 20, true`
B. `true, 20, false`
C. `false, 10, true`
D. `false, 20, false`
E. The code will not compile because of line 5.
F. None of the above.

wolf=false devuelve false, teeth != 10 es true, después true ^ false da como resultado true

5. Which of the following operators are ranked in increasing or the same order of precedence? Assume the + operator is binary addition, not the unary form. (Choose all that apply.)

A. +, *, %, --
B. ++, (int), *
C. =, ==, !
D. (short), =, !, *
E. *, /, %, +, ==
F. !, ||, &
G. ^, +, =, +=

A y C muestran los operadores ordenados

6. What is the output of the following program?

```
1: public class CandyCounter {
2:    static long addCandy(double fruit, float vegetables) {
3:       return (int)fruit+vegetables;
4:    }
5:
6:    public static void main(String[] args) {
7:       System.out.print(addCandy(1.4, 2.4f) + ", ");
8:       System.out.print(addCandy(1.9, (float)4) + ", ");
9:       System.out.print(addCandy((long)(int)(short)2, (float)4)); } }
```

A. `4, 6, 6.0`
B. `3, 5, 6`
C. `3, 6, 6`
D. `4, 5, 6`
E. The code does not compile because of line 9.
F. None of the above.

Error de compilación, el cast (int) se aplica solo a fruit, y el resultado da un float

7. What is the output of the following code snippet?

```
int ph = 7, vis = 2;
boolean clear = vis > 1 & (vis < 9 || ph < 2);
boolean safe = (vis > 2) && (ph++ > 1);
boolean tasty = 7 <= --ph;
System.out.println(clear + "-" + safe + "-" + tasty);
```

A. `true-true-true`
B. `true-true-false`
C. `true-false-true`
D. `true-false-false`
E. `false-true-true`
F. `false-true-false`
G. `false-false-true`
H. `false-false-false`

clear es true, pero en safe se evita el incremento de ph, dejándolo en 7, lo que hace que tasty sea false

8. What is the output of the following code snippet?

```
4: int pig = (short)4;
5: pig = pig++;
6: long goat = (int)2;
7: goat -= 1.0;
8: System.out.print(pig + " - " + goat);
```

A. `4 - 1`
B. `4 - 2`
C. `5 - 1`
D. `5 - 2`
E. The code does not compile due to line 7.
F. None of the above.

pig++ asigna el cuatro antes de incrementar, y goat -= 1.0 le resta 1

9. What are the unique outputs of the following code snippet? (Choose all that apply.)

```
int a = 2, b = 4, c = 2;
System.out.println(a > 2 ? --c : b++);
System.out.println(b = (a!=c ? a : b++));
System.out.println(a > b ? b < c ? b : 2 : 1);
```

A. 1
B. 2
C. 3
D. 4
E. 5
F. 6
G. The code does not compile.

Imprime 4 , luego 5 (se actualiza b), y 1 porque la primera condición falsa

10. What are the unique outputs of the following code snippet? (Choose all that apply.)

```
short height = 1, weight = 3;
short zebra = (byte) weight * (byte) height;
double ox = 1 + height * 2 + weight;
long giraffe = 1 + 9 % height + 1;
System.out.println(zebra);
System.out.println(ox);
System.out.println(giraffe);
```

A. 1
B. 2
C. 3
D. 4
E. 5
F. 6
G. The code does not compile.

Error de compilación porque multiplicar byte * byte se pasa a int, y se debe castear para que quepa en short

11. What is the output of the following code?

```
11: int sample1 = (2 * 4) % 3;
12: int sample2 = 3 * 2 % 3;
13: int sample3 = 5 * (1 % 2);
14: System.out.println(sample1 + ", " + sample2 + ", " + sample3);
```

A. 0, 0, 5
B. 1, 2, 10
C. 2, 1, 5
D. 2, 0, 5
E. 3, 1, 10
F. 3, 2, 6
G. The code does not compile.

Respetando la precedencia de izquierda a derecha: 8%3 es 2, 6%3 es 0, y 5*1 es 5

12. The ________ operator increases a value and returns the original value, while the _______ operator decreases a value and returns the new value.

A. post-increment, post-increment
B. pre-decrement, post-decrement
C. post-increment, post-decrement
D. post-increment, pre-decrement
E. pre-increment, pre-decrement
F. pre-increment, post-decrement

El post-incremento sube el valor pero devuelve el original, y el pre-decremento baja el valor y devuelve el nuevo

13. What is the output of the following code snippet?

```
boolean sunny = true, raining = false, sunday = true;
boolean goingToTheStore = sunny & raining ^ sunday;
boolean goingToTheZoo = sunday && !raining;
boolean stayingHome = !(goingToTheStore && goingToTheZoo);
System.out.println(goingToTheStore + "-" + goingToTheZoo
   + "-" +stayingHome);
```

A. `true-false-false`

B. `false-true-false`

C. `true-true-true`

D. `false-true-true`

E. `false-false-false`

F. `true-true-false`

G. None of the above

13 .goingToTheStore es true , goingToTheZoo es true
y su negación conjunta en stayingHome es false

14. == y != comparan objetos,
la asignación devuelve el valor asignado,
y ! aplica solo a booleanos no a números

14. Which of the following statements are correct? (Choose all that apply.)

A. The return value of an assignment operation expression can be `void`.

B. The inequality operator (`!=`) can be used to compare objects.

C. The equality operator (`==`) can be used to compare a `boolean` value with a numeric value.

D. During runtime, the & and | operators may cause only the left side of the expression to be evaluated.

E. The return value of an assignment operation expression is the value of the newly assigned variable.

F. In Java, `0` and `false` may be used interchangeably.

G. The logical complement operator (`!`) cannot be used to flip numeric values.

15. Which operators take three operands or values? (Choose all that apply.)

A. `=`

B. `&&`

C. `*=`

D. `? :`

E. `&`

F. `++`

G. `/`

El operador condicional ternario (? :) es el único en Java que requiere tres operandos

16. How many lines of the following code contain compiler errors?

```
int note = 1 * 2 + (long)3;
short melody = (byte)(double)(note *= 2);
double song = melody;
float symphony = (float)((song == 1_000f) ? song * 2L : song);
```

A. 0
B. 1
C. 2
D. 3
E. 4

Solo la primera línea falla porque devuelve un long y no se puede asignar a un int

17. Given the following code snippet, what are the values of the variables after it is executed? (Choose all that apply.)

```
int ticketsTaken = 1;
int ticketsSold = 3;
ticketsSold += 1 + ticketsTaken++;
ticketsTaken *= 2;
ticketsSold += (long)1;
```

ticketsSold queda en 5 por el post-incremento y luego suma 1 quedando en 6

A. `ticketsSold` is 8.
B. `ticketsTaken` is 2.
C. `ticketsSold` is 6.
D. `ticketsTaken` is 6.
E. `ticketsSold` is 7.
F. `ticketsTaken` is 4.
G. The code does not compile.

18. Which of the following can be used to change the order of operation in an expression? (Choose all that apply.)

A. `[ ]`
B. `< >`
C. `( )`
D. `\ /`
E. `{ }`
F. `" "`

Solo los paréntesis pueden alterar el orden lógico

19. What is the result of executing the following code snippet? (Choose all that apply.)

```
3: int start = 7;
4: int end = 4;
5: end += ++start;
6: start = (byte)(Byte.MAX_VALUE + 1);
```

A. `start` is `0`.

B. `start` is `-128`.

end suma el pre-incremento de start (8) dando 12,
y al sumar 1 al máximo valor de un byte (127) da un desbordamiento a -128

C. `start` is `127`.

D. `end` is `8`.

E. `end` is `11`.

F. `end` is `12`.

G. The code does not compile.

H. The code compiles but throws an exception at runtime.

20. Which of the following statements about unary operators are true? (Choose all that apply.)

A. Unary operators are always executed before any surrounding numeric binary or ternary operators.

B. The `-` operator can be used to flip a `boolean` value.

C. The pre-increment operator (`++`) returns the value of the variable before the increment is applied.

D. The post-decrement operator (`--`) returns the value of the variable before the decrement is applied.

E. The `!` operator cannot be used on numeric values.

F. None of the above

Los unarios tienen la mayor prioridad,
el post-decremento devuelve el valor original,
! no se usa en tipos numéricos

21. What is the result of executing the following code snippet?

```
int myFavoriteNumber = 8;
int bird = ~myFavoriteNumber;
int plane = -myFavoriteNumber;
var superman = bird == plane ? 5 : 10;
System.out.println(bird + "," + plane + "," + --superman);
```

A. `-7,-8,9`

B. `-7,-8,10`

C. `-8,-8,4`

D. `-8,-8,5`

E. `-9,-8,9`

F. `-9,-8,10`

G. None of the above

bird= ~8 es -9
plane= -8 es -8,
como bird y plane son distintos, superman vale 10,
y al imprimirlo pre-decrementado da 9

Review Questions

The answers to the chapter review questions can be found in the Appendix.

1. Which of the following data types can be used in a `switch` expression? (Choose all that apply.)
   - **A.** `enum`
   - **B.** `int`
   - **C.** `Byte`
   - **D.** `long`
   - **E.** `String`
   - **F.** `char`
   - **G.** `var`
   - **H.** `double`

Los switch aceptan primitivos enteros y sus wrappers, string, enum, y var si corresponde a uno de esos tipos

2. What is the output of the following code snippet? (Choose all that apply.)

```
3: int temperature = 4;
4: long humidity = -temperature + temperature * 3;
5: if (temperature>=4)
6: if (humidity < 6) System.out.println("Too Low");
7: else System.out.println("Just Right");
8: else System.out.println("Too High");
```

   - **A.** `Too Low`
   - **B.** `Just Right`
   - **C.** `Too High`
   - **D.** A `NullPointerException` is thrown at runtime.
   - **E.** The code will not compile because of line 7.
   - **F.** The code will not compile because of line 8.

humidity vale 8
el primer if es verdadero, el segundo falso,
por lo que imprime el primer else

3. Which of the following data types are permitted on the right side of a for-each expression? (Choose all that apply.)
   - **A.** `Double[][]`
   - **B.** `Object`
   - **C.** `Map`
   - **D.** `List`
   - **E.** `String`
   - **F.** `char[]`
   - **G.** `Exception`
   - **H.** `Set`

Un for-each acepta arreglos y clases que implementen Iterable como List o Set

4. What is the output of calling `printReptile(6)`?

```
void printReptile(int category) {
  var type = switch(category) {
      case 1,2 -> "Snake";
      case 3,4 -> "Lizard";
      case 5,6 -> "Turtle";
      case 7,8 -> "Alligator";
   };
   System.out.print(type);
}
```

A. `Snake`

B. `Lizard`

Error de compilación porque el switch no cubre todos los enteros posibles ni incluye un default obligatorio

C. `Turtle`

D. `Alligator`

E. `TurtleAlligator`

F. None of the above

5. What is the output of the following code snippet?

```
List<Integer> myFavoriteNumbers = new ArrayList<>();
myFavoriteNumbers.add(10);
myFavoriteNumbers.add(14);
for (var a : myFavoriteNumbers) {
   System.out.print(a + ", ");
   break;
}

for (int b : myFavoriteNumbers) {
   continue;
   System.out.print(b + ", ");
}

for (Object c : myFavoriteNumbers)
   System.out.print(c + ", ");
```

A. It compiles and runs without issue but does not produce any output.

B. `10, 14,`

C. `10, 10, 14,`

Hay un error de compilación porque el sysout del segundo for-each es inalcanzable por el continue

D. `10, 10, 14, 10, 14,`

E. Exactly one line of code does not compile.

F. Exactly two lines of code do not compile.

G. Three or more lines of code do not compile.

H. The code contains an infinite loop and does not terminate.

6. El bucle for evalúa su condición antes de,
un switch expression requiere default si retorna un valor,
y el do-while siempre corre al menos una vez

6. Which statements about decision structures are true? (Choose all that apply.)

   A. A for-each loop can be executed on any Collections Framework object.
   B. The body of a `while` loop is guaranteed to be executed at least once.
   C. The conditional expression of a `for` loop is evaluated before the first execution of the loop body.
   D. A `switch` expression that takes a `String` and assigns the result to a variable requires a `default` branch.
   E. The body of a `do/while` loop is guaranteed to be executed at least once.
   F. An `if` statement can have multiple corresponding `else` statements.

7. Assuming `weather` is a well-formed nonempty array, which code snippet, when inserted independently into the blank in the following code, prints all of the elements of `weather`? (Choose all that apply.)

```
private void print(int[] weather) {
   for(____________________) {
      System.out.println(weather[i]);
   }
}
```

Son las únicas correctas para recorrer el arreglo sin salir de los límites

   A. `int i=weather.length; i>0; i--`
   B. `int i=0; i<=weather.length-1; ++i`
   C. `var w : weather`
   D. `int i=weather.length-1; i>=0; i--`
   E. `int i=0, int j=3; i<weather.length; ++i`
   F. `int i=0; ++i<10 && i<weather.length;`
   G. None of the above

8. What is the output of calling `printType(11)`?

```
void printType(Object o) {
   if(o instanceof Integer bat) {
      System.out.print("int");
   } else if(o instanceof Integer bat && bat < 10) {
      System.out.print("small int");
   } else if(o instanceof Long bat || bat <= 20) {
      System.out.print("long");
   } default {
      System.out.print("unknown");
   }
}
```

A. `int`

B. `small int`

Error de compilación, bat sale de scope en la línea 36 porque no es Long, default no es una palabra reservada válida en un bloque if/else

C. `long`

D. `unknown`

E. Nothing is printed.

F. The code contains one line that does not compile.

G. The code contains two lines that do not compile.

H. None of the above

9. Which statements, when inserted independently into the following blank, will cause the code to print 2 at runtime? (Choose all that apply.)

```
int count = 0;
BUNNY: for(int row = 1; row <=3; row++)
   RABBIT: for(int col = 0; col <3 ; col++) {
      if((col + row) % 2 == 0)
         ____________________;
      count++;
   }
System.out.println(count);
```

Al escapar prematuramente del bucle con los break o continue se evita que count suba más allá de 2

A. `break BUNNY`

B. `break RABBIT`

C. `continue BUNNY`

D. `continue RABBIT`

E. `break`

F. `continue`

G. None of the above, as the code contains a compiler error.

10. Given the following method, how many lines contain compilation errors? (Choose all that apply.)

```
private DayOfWeek getWeekDay(int day, final int thursday) {
   int otherDay = day;
   int Sunday = 0;
   switch(otherDay) {
      default:
      case 1: continue;
      case thursday: return DayOfWeek.THURSDAY;
      case 2,10: break;
```

```
         case Sunday: return DayOfWeek.SUNDAY;
         case DayOfWeek.MONDAY: return DayOfWeek.MONDAY;
      }
      return DayOfWeek.FRIDAY;
   }
```

A. None, the code compiles without issue.

B. 1

C. 2

D. 3

E. 4

F. 5

G. 6

H. The code compiles but may produce an error at runtime.

Cuatro líneas fallan
un continue ilegal en un switch
casos usando variables no constantes
y un caso comparando con un tipo distinto a int

11. What is the output of calling `printLocation(Animal.MAMMAL)`?

```
10: class Zoo {
11:    enum Animal {BIRD, FISH, MAMMAL}
12:    void printLocation(Animal a) {
13:       long type = switch(a) {
14:          case BIRD -> 1;
15:          case FISH -> 2;
16:          case MAMMAL -> 3;
17:          default -> 4;
18:       };
19:       System.out.print(type);
20:    } }
```

A. 3

B. 4

C. 34

D. The code does not compile because of line 13.

E. The code does not compile because of line 17.

F. None of the above

Coincide con MAMMAL devolviendo 3

12. What is the result of the following code snippet?

```
int sing = 8, squawk = 2, notes = 0;
while(sing > squawk) {
   sing--;
   squawk += 2;
```

```
7:    notes += sing + squawk;
8: }
9: System.out.println(notes);
```

El ciclo itera dos veces
notes suma primero 11 (7+4)
luego suma 12 (6+6),
dando 23

A. 11

B. 13

C. 23

D. 33

E. 50

F. The code will not compile because of line 7.

13. What is the output of the following code snippet?

```
2: boolean keepGoing = true;
3: int result = 15, meters = 10;
4: do {
5:    meters--;
6:    if(meters==8) keepGoing = false;
7:    result -= 2;
8: } while keepGoing;
9: System.out.println(result);
```

Error de compilación, la expresión del do-while (línea 8) no tiene los paréntesis obligatorios

A. 7

B. 9

C. 10

D. 11

E. 15

F. The code will not compile because of line 6.

G. The code does not compile for a different reason.

14. Which statements about the following code snippet are correct? (Choose all that apply.)

```
for(var penguin : new int[2])
   System.out.println(penguin);
var ostrich = new Character[3];
for(var emu : ostrich)
   System.out.println(emu);
List<Integer> parrots = new ArrayList<Integer>();
for(var macaw  : parrots)
   System.out.println(macaw);
```

var extrae los tipos
int para el array primitivo,
Character para el array de objetos,
e Integer para la lista

A. The data type of `penguin` is `Integer`.

B. The data type of `penguin` is `int`.

C. The data type of `emu` is undefined.

D. The data type of `emu` is `Character`.

E. The data type of `macaw` is `List`.

F. The data type of `macaw` is `Integer`.

G. None of the above, as the code does not compile.

15. What is the result of the following code snippet?

```
final char a = 'A', e = 'E';
char grade = 'B';
switch (grade) {
   default:
   case a:
   case 'B': 'C': System.out.print("great ");
   case 'D': System.out.print("good "); break;
   case e:
   case 'F': System.out.print("not good ");
}
```

Error de compilación,
no se pueden encadenar casos usando dos en un case case 'B': 'C':

A. `great`

B. `great good`

C. `good`

D. `not good`

E. The code does not compile because the data type of one or more `case` statements does not match the data type of the `switch` variable.

F. None of the above

16. Given the following array, which code snippets print the elements in reverse order from how they are declared? (Choose all that apply.)

```
char[] wolf = {'W', 'e', 'b', 'b', 'y'};
```

A.

```
int q = wolf.length;
for( ; ; ) {
   System.out.print(wolf[--q]);
   if(q==0) break;
}
```

En estas opciones los índices imprimen
desde el último carácter hasta el primero

B.

```
for(int m=wolf.length-1; m>=0; --m)
   System.out.print(wolf[m]);
```

C.

```
for(int z=0; z<wolf.length; z++)
   System.out.print(wolf[wolf.length-z]);
```

D.

```
int x = wolf.length-1;
for(int j=0; x>=0 && j==0; x--)
   System.out.print(wolf[x]);
```

E.

```
final int r = wolf.length;
for(int w = r-1; r>-1; w = r-1)
   System.out.print(wolf[w]);
```

F.

```
for(int i=wolf.length; i>0; --i)
   System.out.print(wolf[i]);
```

G. None of the above

17. What distinct numbers are printed when the following method is executed? (Choose all that apply.)

```
private void countAttendees() {
   int participants = 4, animals = 2, performers = -1;
   while((participants = participants+1) < 10) {}
   do {} while (animals++ <= 1);
   for( ; performers<2; performers+=2) {}

   System.out.println(participants);
   System.out.println(animals);
   System.out.println(performers);
}
```

El primer ciclo detiene a participants en 10,
y el segundo ciclo do-while se ejecuta dejando a animals en 3

A. 6

B. 3

C. 4

D. 5

E. 10

F. 9

G. The code does not compile.

H. None of the above

16. El pattern matching utiliza instanceof y flow scoping previene que uses la variable si el cast falla

18. Which statements about pattern matching and flow scoping are correct? (Choose all that apply.)

A. Pattern matching with an `if` statement is implemented using the `instance` operator.

B. Pattern matching with an `if` statement is implemented using the `instanceon` operator.

C. Pattern matching with an `if` statement is implemented using the `instanceof` operator.

D. The pattern variable cannot be accessed after the `if` statement in which it is declared.

E. Flow scoping means a pattern variable is only accessible if the compiler can discern its type.

F. Pattern matching can be used to declare a variable with an `else` statement.

19. What is the output of the following code snippet?

```
double iguana = 0;
do {
   int snake = 1;
   System.out.print(snake++ + " ");
   iguana--;
} while (snake <= 5);
System.out.println(iguana);
```

Error de compilación
la variable snake se declara dentro del bloque do,
por lo que no existe en el while

A. `1 2 3 4 -4.0`

B. `1 2 3 4 -5.0`

C. `1 2 3 4 5 -4.0`

D. `0 1 2 3 4 5 -5.0`

E. The code does not compile.

F. The code compiles but produces an infinite loop at runtime.

G. None of the above

20. Which statements, when inserted into the following blanks, allow the code to compile and run without entering an infinite loop? (Choose all that apply.)

```
int height = 1;
L1: while(height++ <10) {
   long humidity = 12;
   L2: do {
      if(humidity-- % 12 == 0) ____________;
      int temperature = 30;
      L3: for( ; ; ) {
         temperature++;
         if(temperature>50) ____________;
      }
   } while (humidity > 4);
}
```

20 .Las instrucciones con L2 permiten romper o saltar directamente el bucle exterior y evita el bucle infinito interior

A. `break L2` on line 8; `continue L2` on line 12

B. `continue` on line 8; `continue` on line 12

C. `break L3` on line 8; `break L1` on line 12

D. `continue L2` on line 8; `continue L3` on line 12

E. `continue L2` on line 8; `continue L2` on line 12

F. None of the above, as the code contains a compiler error

21. A minimum of how many lines need to be corrected before the following method will compile?

```
21: void findZookeeper(Long id) {
22:    System.out.print(switch(id) {
23:       case 10 -> {"Jane"}
24:       case 20 -> {yield "Lisa";};
25:       case 30 -> "Kelly";
26:       case 30 -> "Sarah";
27:       default -> "Unassigned";
28:    });
29: }
```

Fallan cuatro cosas:
tipo Long inválido en switch
falta punto y coma,
sobra un punto y coma en una llave,
y hay un case 30 duplicado

A. Zero

B. One

C. Two

D. Three

E. Four

F. Five

22. What is the output of the following code snippet? (Choose all that apply.)

```
2: var tailFeathers = 3;
3: final var one = 1;
4: switch (tailFeathers) {
5:    case one: System.out.print(3 + " ");
6:    default: case 3: System.out.print(5 + " ");
7: }
8: while (tailFeathers > 1) {
9:    System.out.print(--tailFeathers + " "); }
```

A. `3`

B. `5 1`

C. `5 2`

Al buscar el 3 cae en el default case 3 imprimiendo 5
luego el bucle while lo reduce progresivamente imprimiendo 2 y 1

D. `3 5 1`

E. `5 2 1`

F. The code will not compile because of lines 3–5.

G. The code will not compile because of line 6.

23. What is the output of the following code snippet?

```
15: int penguin = 50, turtle = 75;
16: boolean older = penguin >= turtle;
17: if (older = true) System.out.println("Success");
18: else System.out.println("Failure");
19: else if(penguin != 50) System.out.println("Other");
```

A. `Success`
B. `Failure`
C. `Other`
D. The code will not compile because of line 17.
E. The code compiles but throws an exception at runtime.
F. None of the above

Error de compilación en el último else if ya que esta no pertenece a la estructura if-else anterior

24. Which of the following are possible data types for `friends` that would allow the code to compile? (Choose all that apply.)

```
for(var friend in friends) {
   System.out.println(friend);
}
```

A. `Set`
B. `Map`
C. `String`
D. `int[]`
E. `Collection`
F. `StringBuilder`
G. None of the above

Error de compilación porque el for-each en Java usa dos puntos no la palabra “in”

25. What is the output of the following code snippet?

```
String instrument = "violin";
final String CELLO = "cello";
String viola = "viola";
int p = -1;
switch(instrument) {
   case "bass" : break;
   case CELLO : p++;
   default: p++;
   case "VIOLIN": p++;
   case "viola" : ++p; break;
}
System.out.print(p);
```

A. -1

B. 0

C. 1

D. 2

E. 3

F. The code does not compile.

Como "violin" en minúsculas no coincide con nada,
se ejecuta el default, pero como no tiene break cae en los siguientes casos
e incrementa p tres veces hasta 2

26. What is the output of the following code snippet? (Choose all that apply.)

```
9:  int w = 0, r = 1;
10: String name = "";
11: while(w < 2) {
12:    name += "A";
13:    do {
14:       name += "B";
15:       if(name.length()>0) name += "C";
16:       else break;
17:    } while (r <=1);
18:    r++; w++; }
19: System.out.println(name);
```

A. ABC

B. ABCABC

C. ABCABCABC

D. Line 15 contains a compilation error.

E. Line 18 contains a compilation error.

F. The code compiles but never terminates at runtime.

G. The code compiles but throws a `NullPointerException` at runtime.

Genera un bucle infinito en el entorno de ejecución
porque la variable r del do-while anidado solo se actualiza fuera de este

27. What is printed by the following code snippet?

```
byte amphibian = 1;
String name = "Frog";
String color = switch(amphibian) {
   case 1 -> { yield "Red"; }
   case 2 -> { if(name.equals("Frog")) yield "Green"; }
   case 3 -> { yield "Purple"; }
   default -> throw new RuntimeException();
};
System.out.print(color);
```

Error de compilación el caso 2 no garantiza devolver un valor yield , esto rompe la regla del switch

A. `Red`

B. `Green`

C. `Purple`

D. `RedPurple`

E. An exception is thrown at runtime.

F. The code does not compile.

28. What is the output of calling `getFish("goldie")`?

```
void getFish(Object fish) {
   if (!(fish instanceof String guppy))
      System.out.print("Eat!");
   else if (!(fish instanceof String guppy)) {
      throw new RuntimeException();
   }
   System.out.print("Swim!");
}
```

Error de compilación porque se declaran dos variables con el mismo nombre guppy

A. `Eat!`

B. `Swim!`

C. `Eat!` followed by an exception.

D. `Eat!Swim!`

E. An exception is printed.

F. None of the above

29. What is the result of the following code?

```
1: public class PrintIntegers {
2:    public static void main(String[] args) {
3:       int y = -2;
4:       do System.out.print(++y + " ");
5:       while(y <= 5);
6: } }
```

Comienza imprimiendo desde -1 por el pre-incremento, y se detiene justo después de imprimir el 6

A. `-2 -1 0 1 2 3 4 5`

B. `-2 -1 0 1 2 3 4`

C. `-1 0 1 2 3 4 5 6`

D. `-1 0 1 2 3 4 5`

E. The code will not compile because of line 5.

F. The code contains an infinite loop and does not terminate.